

Exercise 3

$X \rightarrow \text{C.R.V.}$

$$\Omega_X = \mathbb{R}$$

And $Y \rightarrow \text{R.V.}$

$$X = \ln Y$$

① Ω_Y ?

② $F_Y(y)$? (wrt $F_X(x)$)

③ $f_Y(y)$? (wrt $f_X(\cdot)$)

We can assume that X is gaussian
 $X \sim N(\mu, \sigma^2)$

④ $E[Y]$?

Lets define the median value m of X as

$$P[X \geq m] \geq \frac{1}{2} \quad \& \quad P[X \leq m] \geq \frac{1}{2}$$

⑥ Deduce m_x & m_y

$$\Rightarrow X = \ln Y \quad Y = e^X$$

$$(1) \quad \Omega_Y = 0, \infty[$$

$$(2) \quad F_Y(y) = P[Y \leq y]$$

$$= P[e^X \leq y]$$

$$= P[X \leq \ln y]$$

$$= \underline{\underline{F_X[\ln y]}}$$

$$\textcircled{3} \quad f_Y(y) = \frac{d}{dy} (F_Y(y))$$

$$= \frac{d}{dy} F_X(\ln y)$$

$$\Rightarrow f'_X(\ln y) \cdot \frac{1}{y} \quad y = e^x$$

$$\Rightarrow \frac{f'_X(x)}{e^x}$$

$$\textcircled{4} \quad E[Y] = \int y \cdot f_Y(y)$$

$$X \sim N(\mu, \sigma^2)$$

$$E[Y]$$

$$\mu^2 + a^2\sigma^4 + 2a\sigma^2\mu$$

$$\Rightarrow \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} e^{ax - \frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$\Rightarrow \frac{x^2 - 2(\mu + a\sigma^2)x + \mu^2}{2\sigma^2}$$

$$\Rightarrow x^2 - 2(\mu + a\sigma^2)x + \mu^2 + a^2\sigma^4 + 2a\sigma^2\mu - a^2\sigma^4 - 2a\sigma^2\mu$$

$$\Rightarrow \frac{(x - (\mu + a\sigma^2))^2 - a^2\sigma^4 - 2a\sigma^2\mu}{2\sigma^2}$$

$$\Rightarrow \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} \frac{e^{-\left(\frac{a^2\sigma^4 + 2a\sigma^2\mu}{2\sigma^2}\right)} e^{-\frac{(x - (\mu + a\sigma^2))^2}{2\sigma^2}} dx}{\sqrt{2\pi\sigma^2}}$$

Another normal distribution &
hence $\int_{-\infty}^{\infty} \dots = 1$

from normality
axiom.

$$\Rightarrow \frac{e^{-\left(\frac{a^2\sigma^4 + 2a\sigma^2\mu}{2\sigma^2}\right)}}{\sqrt{2\pi\sigma^2}} = \underline{\underline{E[y^a]}}$$

$$\Rightarrow f_Z(z) = \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{\sqrt{1+z^2}} \cdot e^{-\frac{(\ln(1+z^2))^2}{2z^2}}$$

$$\lim_{\sigma \rightarrow 0} f_Z(z) \Rightarrow \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{1} \cdot e^{-\frac{(\ln(1+z^2))^2}{2z^2}}$$

↳ using $\lim_{x \rightarrow 0} \ln(1+x) = x$

$$\Rightarrow \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{z^2 \cancel{\sigma^2}}{2\cancel{\sigma^2}}}$$

$$\Rightarrow \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

↳ $N(0, 1)$

ADVANCED TECHNIQUES

1. Transform

The transform associated with R.V. X .
↓
AKA
↓
moment generating function.

$$\cdot M_X(s) = E[e^{sx}]$$

⇒ D.R.V

$$M_X(s) = \sum_{x \in \mathcal{R}_X} e^{sx} \cdot P_X(x) dx$$

⇒ C.R.V

$$M_X(s) = \int f_X(x) \cdot e^{sx} \cdot dx$$

$M_x(s) \rightarrow$ Laplace Transformation
Fourier.

From Transform to Moments

$$M_x(s) = \int e^{sx} \cdot f(x) dx$$

$$\frac{\partial M_x(s)}{\partial s} = \int x \cdot e^{sx} \cdot f(x) dx$$

$$\left. \frac{\partial M_x(s)}{\partial s} \right|_{s=0} = E[x]$$

$$\Rightarrow \left. \frac{\partial^n M_x(s)}{\partial s^n} \right|_{s=0} = E[x^n]$$

Sums of independent R.V.

$$W = X + Y$$

P.D.F of W can be obtained by computing & inverting the transform.

$$\begin{aligned}\Rightarrow M_W(s) &= E[e^{sW}] = E[e^{s(X+Y)}] \\ &= E[e^{sX} \cdot e^{sY}]\end{aligned}$$

Since x & y are independent

$$M_W(s) = E[e^{sX}] \cdot E[e^{sY}] = M_X(s) \cdot M_Y(s)$$

$$M_W(s) = M_X(s) \cdot M_Y(s)$$

The discrete case

$$P_w(\omega) = P[X + Y = \omega]$$

$$= \sum_{\{x, y \mid x + y = \omega\}} P[X = x, Y = y]$$

$$= \sum_x P[X = x, Y = \omega - x]$$

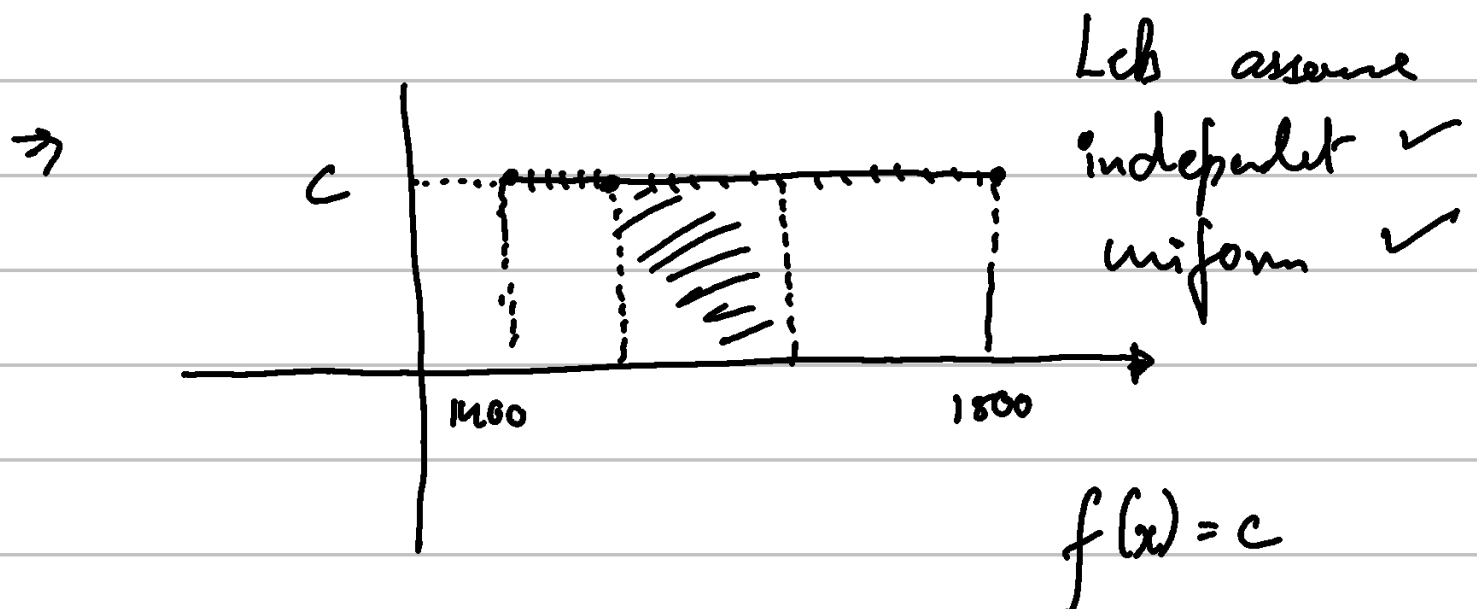
$$= \sum_x P[X = x] \cdot P[Y = \omega - x]$$

Ex 2 friends.

14:00 - 18:00

waits 15 min

probab of meeting



$$P[a \leq x \leq b]$$

$$\int_0^{3600} c \, dx$$

⇒ $P[x = a] \cdot P[a \leq x \leq a+15]$

$$2 \cdot P[X = a] \cdot P[a \leq Y \leq a + 15 \text{ min}]$$

$$2 \cdot \int_a^{a+15 \text{ min}} \frac{1}{60 \text{ min}} \cdot \frac{1}{4} dx$$

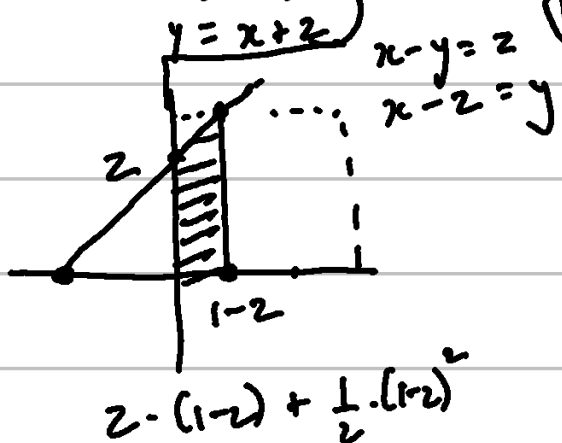
wrong here
limits
are
incorrect.

$$\Rightarrow 2 \cdot \frac{1}{4 \cdot 60 \text{ min}} \cdot \frac{1}{4} \cdot 15 \text{ min}$$

$$\Rightarrow \frac{2}{8} = \frac{1}{4}$$

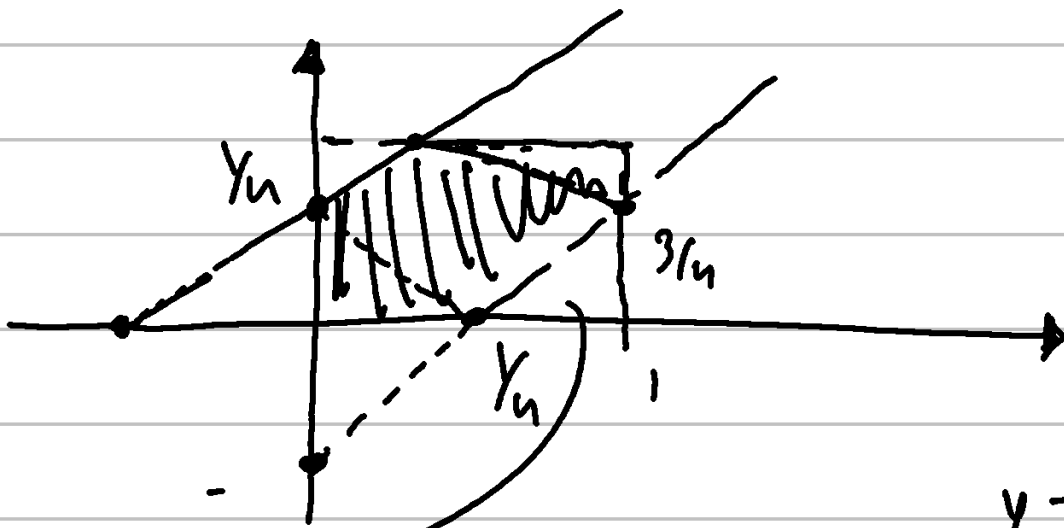
when $a \leq 45 \text{ min}$ after 2.

Modeling it from 0-1 instead of 1400h to 1500h



$$\iint_{\{(x,y) | |x-y| < 2\}} \frac{1}{1-0} \cdot \frac{1}{1-0} dx \cdot dy$$

$$f_{xy}(x, y) = f(x) \cdot f(y)$$



$$y = 1 - 1$$

$$y - x = 2$$

$$y - x = -2$$

$$y = x + 2$$

$$y = x - 2$$

$$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} \Rightarrow \frac{9}{32} \cdot 2 = \frac{9}{16}$$

Area of domain = $1 - \frac{9}{16} = \frac{7}{16}$

Exercise

(X, Y) ind. C.R.V.

$$X \sim e(\lambda)$$

$$Y \sim e(\lambda)$$

$$\underline{\lambda = 1}$$

$$\lambda e^{-\lambda x}$$

$$\underline{\underline{e^{-x}}}$$

$$Z = X + Y$$

① Find the conditional law of Z given x_0 .

$$F_{Z/X=x_0}(z)$$

$$F_Z(z \leq z) = \int_0^z \int_0^z e^{-x} e^{-y} dx dy.$$

$$\boxed{F_Z(z \leq z) = -e^{-x} \Big|_0^z \cdot -e^{-y} \Big|_0^z = (1 - e^{-z})^2}$$

$$\begin{aligned}
 F_{Z/X=X}(z) &= P[Z \leq z / X=x] \\
 &= P[X+Y \leq z] \\
 &= P[Y \leq z-x] \\
 &= F_Y[z-x]
 \end{aligned}$$

$$\begin{aligned}
 f_{Z/X=X}(z) &= f_Y(z-x) \\
 \Rightarrow f_Z(z) &= \int_{-\infty}^{\infty} \underbrace{f_{Z,X}(z,x)}_{f_{Z/X=X}(z) \cdot f_X(x)} dx \\
 &= \int_{-\infty}^{\infty} f_{Z/X=X}(z) \cdot f_X(x) dx \\
 &= \int_0^{\infty} f_Y(z-x) f_X(x) dx
 \end{aligned}$$

